

CHAPTER 1` : COMMUNICATION IN A CONNECTED WORLD

MODULE OVERVIEW

Modern society depends on computer networks for communication, business, education, entertainment, healthcare, and government services. Understanding how devices communicate, how data travels across networks, and how network performance is measured is fundamental for every IT professional.

This module introduces students to the essential concepts of networking, including network components, network types, methods of data transmission, bandwidth, and throughput. Students will also perform hands-on laboratory activities using networking simulation software.

LEARNING OUTCOMES

At the end of this module, students should be able to:

At the end of this lesson, the students will be able to:

1. recognize the importance of communication in today's digital society.
2. explain how networks enable global connectivity.
3. identify real-life applications of communication technologies (e.g., email, social media, video conferencing).
4. recognize the impact of communication technologies on business, education, and daily life.

INTRODUCTION

Every day billions of devices communicate across the Internet. Whether sending an email, watching YouTube, joining Zoom meetings, or using cloud applications, information travels through interconnected computer networks.

Computer networking enables:

- Communication
- Resource sharing
- Remote access
- Cloud computing
- Online education
- Smart devices (IoT)

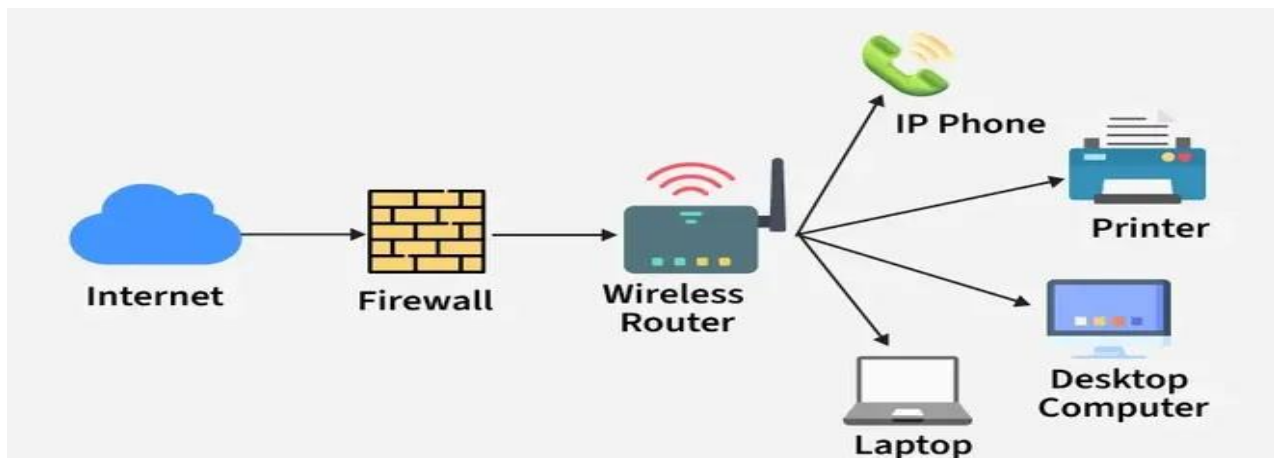
- E-commerce
- Digital banking

Without networking, today's digital world would not exist.

I. NETWORK COMPONENTS, TYPES, AND CONNECTIONS

What is a Computer Network?

A computer network is a collection of interconnected devices that exchange data using communication protocols.



Basic Network Components

1. End Devices (Hosts)

These are devices used by people.

Examples:

- Desktop
- Laptop
- Smartphone
- Tablet
- Printer
- Server
- Smart TV

Functions:

- Send data
- Receive data
- Process information

2. Intermediate Devices

These connect different devices together.

Router



Functions

- Connects different networks
- Selects best path
- Connects LAN to Internet

Example

Home Wi-Fi Router

Switch



Functions

- Connects devices inside LAN
- Uses MAC Address
- Faster communication

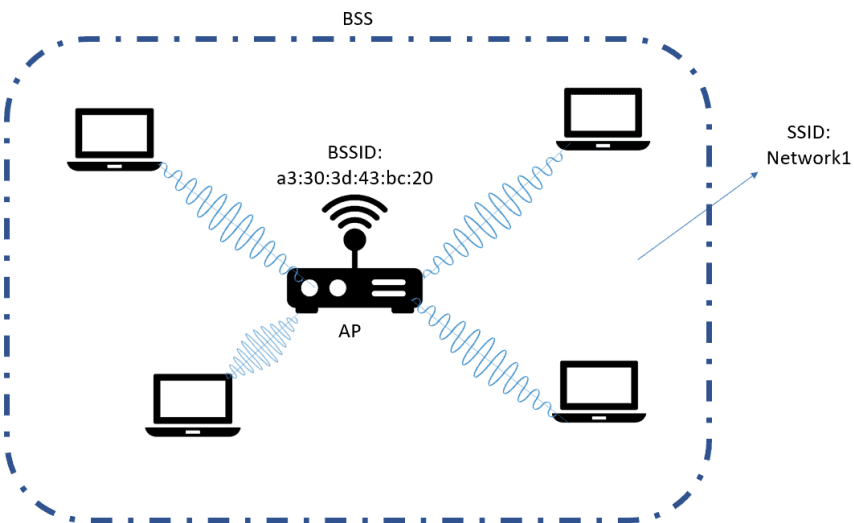
Hub



Functions

- Broadcasts data to all devices
- Older technology
- Less secure

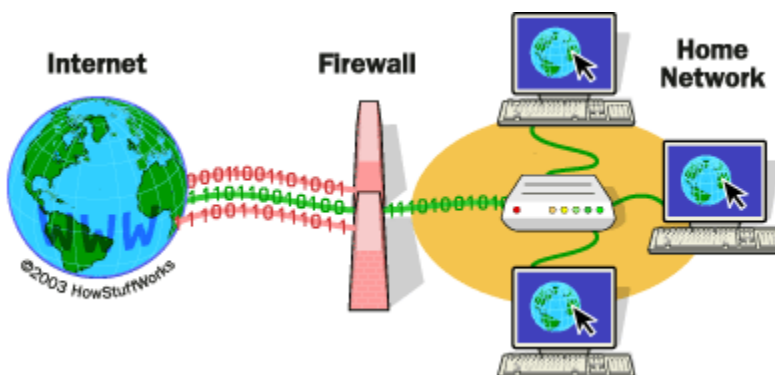
Access Point



Functions

- Provides wireless connection
- Connects Wi-Fi devices

Firewall



Functions

- Filters incoming traffic

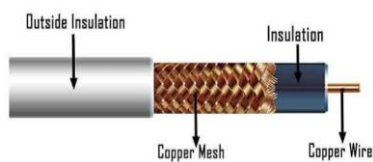
- Blocks unauthorized access
- Improves security

NETWORK MEDIA

Wired

Examples

- UTP Cable
- STP Cable
- Fiber Optic
- Coaxial



Advantages

- Stable
- Faster
- Secure

Disadvantages

- Limited mobility

Wireless

Examples

- Wi-Fi
- Bluetooth
- Satellite



Advantages

- Mobility
- Easy installation

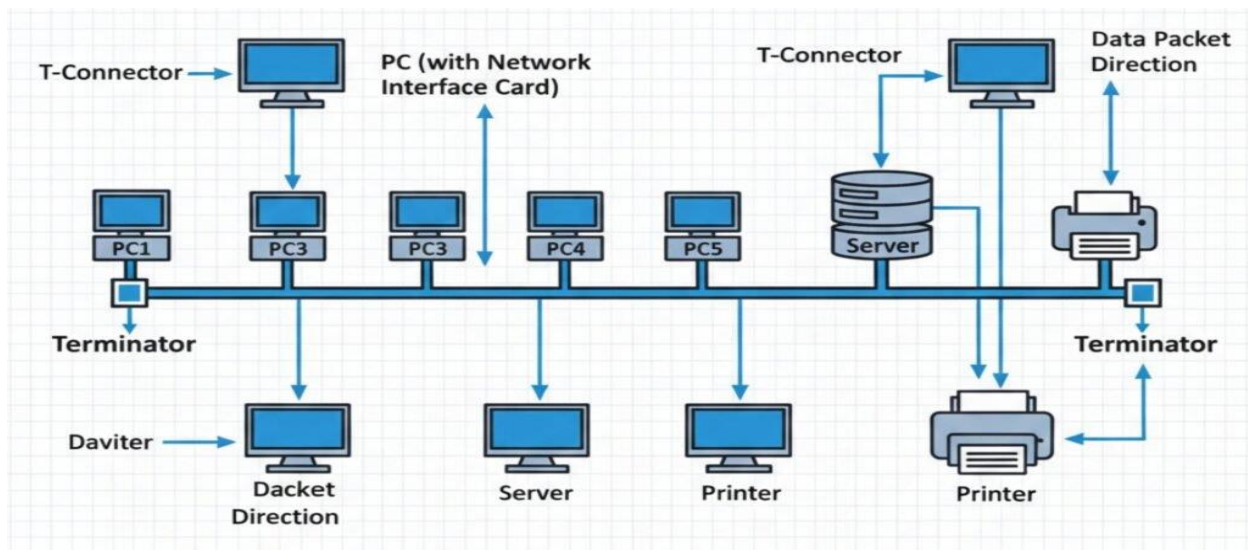
Disadvantages

- Signal interference
- Security concerns

NETWORK TOPOLOGIES

1. Bus

A bus topology is a network layout where every device connects to a single central cable, known as the "backbone" or "bus". Data broadcasts across this shared cable to all nodes, but only the intended recipient processes it. Both ends of the cable require terminators to prevent signal reflection.



Advantages

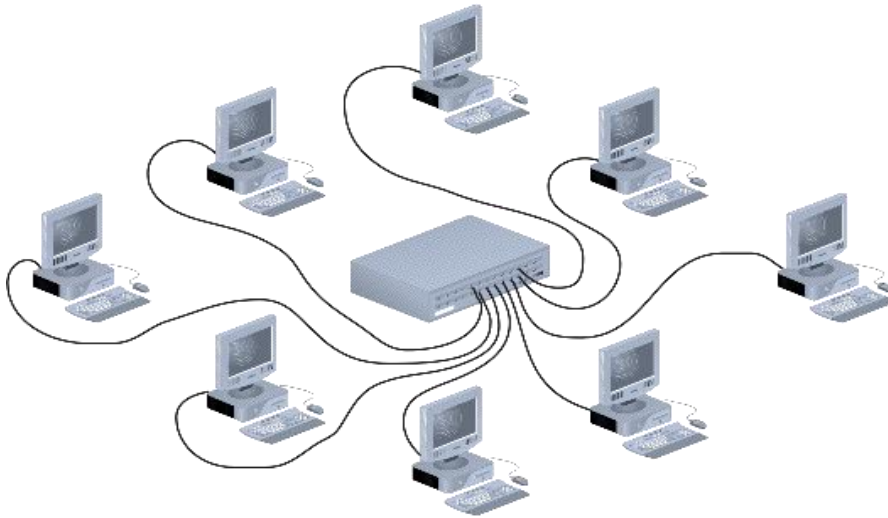
- Cheap
- Easy installation

Disadvantages

- Single cable failure affects network

2. Star

a network setup where every device (computer, printer, etc.) connects individually to a central hub or switch. The central node manages all data traffic. It is the most common layout for modern homes, offices, and Local Area Networks (LANs).



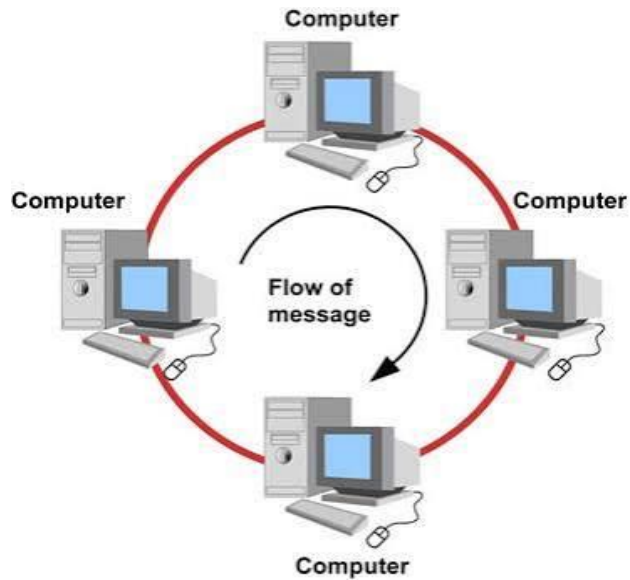
Most common today

Advantages

- Easy troubleshooting
- High performance

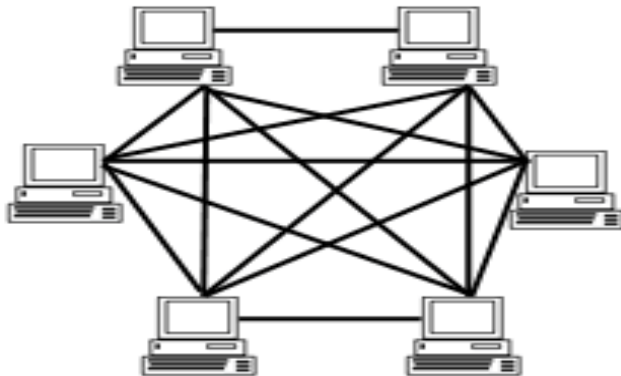
3. Ring

a network setup where devices (nodes) are connected in a circular, closed loop. Each device links to exactly two others. Data generally travels sequentially in one direction, often using a "token" to manage traffic and prevent data collisions. Devices connected in circular pattern



4. Mesh

a network setup where devices (nodes) are interconnected, creating multiple pathways for data to travel. This decentralized structure guarantees high reliability and redundancy, as data can dynamically self-route around failed connections.



Every device connected to multiple devices

Advantages

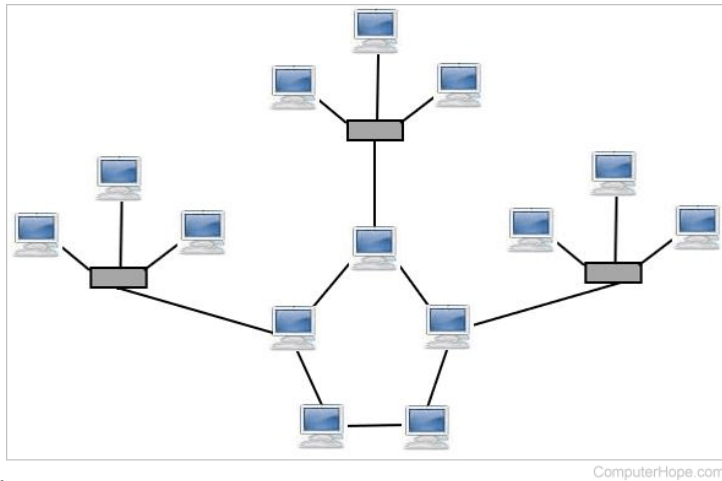
- High reliability

Disadvantages

- Expensive

5. Hybrid

a custom network architecture that interconnects two or more standard topologies (like Star, Bus, or Ring) to fit specific organizational needs. It balances flexibility and fault tolerance, making it ideal for large, growing networks.

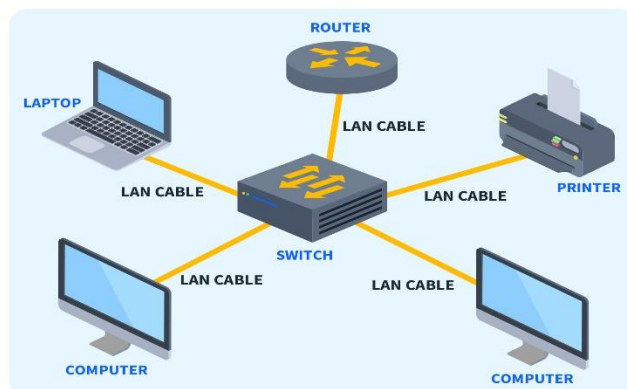


Combination of different topologies

TYPES OF NETWORKS

1. Local Area Network (LAN)

Group of interconnected computers and devices within a limited geographical area, such as a home, school, or office building. It allows devices to communicate, share resources (like files and printers), and access the internet.



Examples

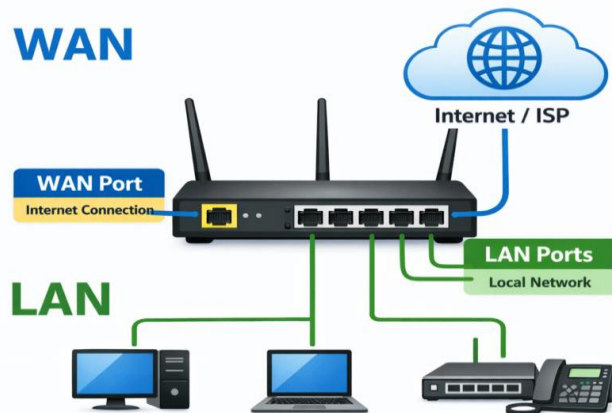
School

Office

Home

2. Wide Area Network (WAN)

any network that extends over a large geographic area, usually connecting multiple local area networks (LANs).



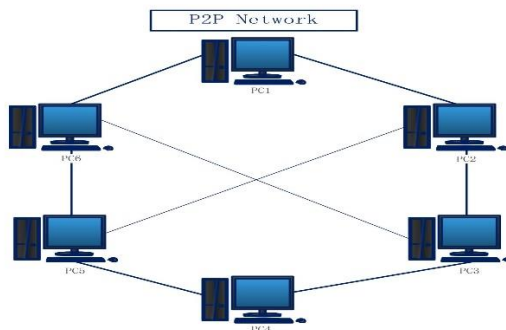
Example

Internet

NETWORK MODELS

1. Peer-to-Peer (P2P)

is a decentralized architecture where each connected device (node) acts as both a client and a server. Instead of relying on a central hub, participants directly share resources like bandwidth, storage, and processing power with one another.



Advantages

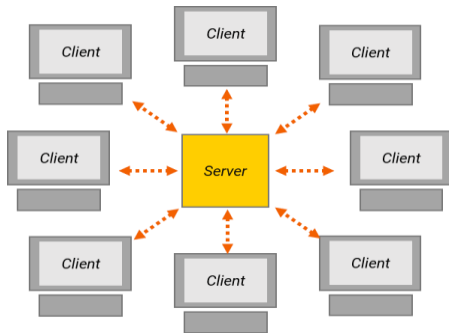
- Cheap
- Easy

Disadvantages

- Weak security

2. Client-Server

a centralized computing model where multiple devices (clients) connect to a central computer (server) to access and share resources. Clients request data or services, and the server processes those requests and returns a response, making it highly efficient for managing security, backups, and shared applications



Advantages

- Centralized
- Secure
- Scalable

Laboratory Activity

Using Cisco Packet Tracer

Create:

- 4 PCs
- 1 Switch
- 1 Router

Configure

- IP Address
- Ping Test

II. DATA TRANSMISSION

What is Data Transmission?

Data transmission is the movement of digital information between devices.

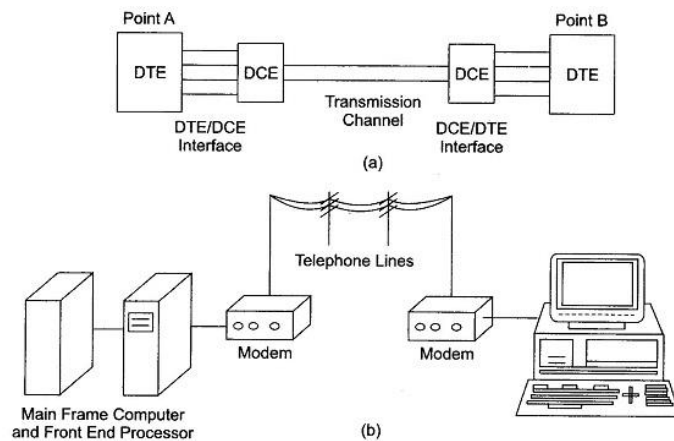


Fig. 18.1 Data Transmission System (Basic)
(a) Block Diagram (b) Pictorial

DATA REPRESENTATION

Data is represented as:

- **Binary** - converting between standard numbers (decimal) and the binary system (base-2, using only 0s and 1s)
- **Bits** - the smallest unit of data. It can hold only one of two values: 0 or 1. Bits form the foundation of all digital information, and are often grouped into sets of eight to form a byte.
- **Bytes** - the basic unit of digital information in computing, traditionally composed of 8 bits.

TRANSMISSION MEDIA

Guided

- Twisted Pair
- Fiber
- Coaxial

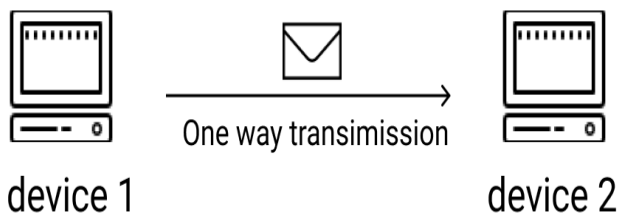
Unguided

- Radio
- Infrared
- Microwave
- Satellite

DATA TRANSMISSION MODES

Simplex

One-way communication

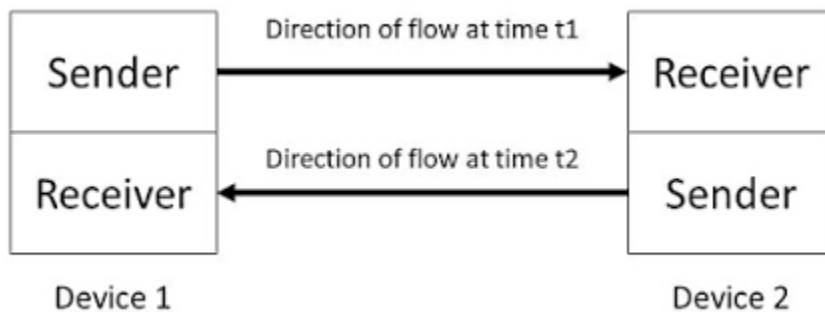


Example

Television

Half Duplex

Both directions but one at a time

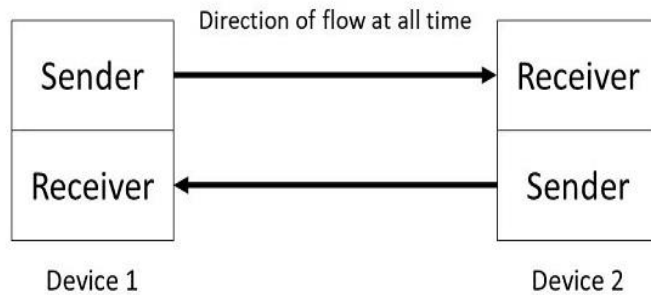


Example

Walkie-talkie

Full Duplex

Both directions simultaneously



Example

Telephone

TRANSMISSION TYPES**Serial Transmission**

sends data bits sequentially, one at a time, over a single wire or communication channel. It is highly cost-effective and reliable over long distances compared to parallel transmission, making it the industry standard for modern networking, USB connections, and integrated circuits.

One bit at a time



Advantages

- Reliable

Examples

USB

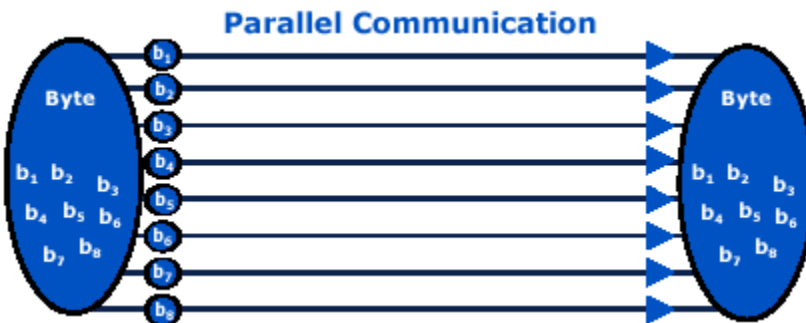
Ethernet

Parallel Transmission

is a data communication method where multiple bits are sent simultaneously over separate wires or channels. While highly fast because an entire byte (8 bits) travels across 8 distinct lines in a single clock cycle, its cost limits it to short distances like inside computers.

Multiple bits simultaneously

Older technology



III. BANDWIDTH AND THROUGHPUT

What is Bandwidth?

Bandwidth refers to the maximum capacity of a communication channel.

Units

bps

Kbps

Mbps

Gbps

Tbps

Example

100 Mbps Internet

Maximum speed

100 Mbps

Throughput

Actual amount of data transferred.

Formula

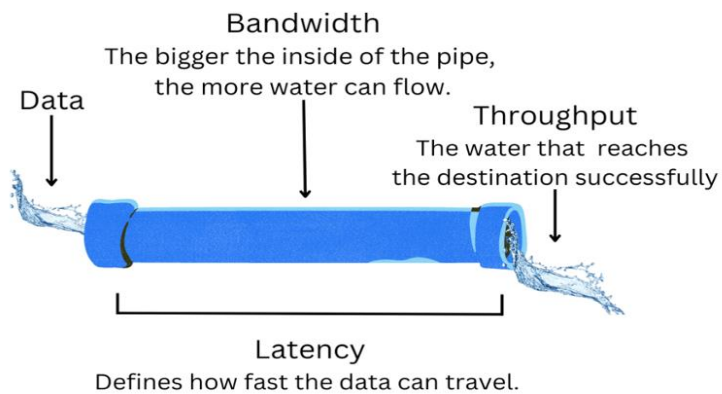
Throughput = Data Successfully Delivered ÷ Time

Example

100 MB downloaded in 20 seconds

Throughput

5 MB/s



Factors Affecting Throughput

Network congestion

Distance

Signal quality

Interference

Hardware capability

Protocols

Errors

Bandwidth